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Explainable artificial intelligence in the talent recruitment process-a literature review

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ABSTRACT

With the widespread application of artificial intelligence in talent recruitment, the 'black-box' nature of AI, which leads to insufficient transparency and interpretability in decision-making, has gradually become a key challenge. This paper reviews the application of Explainable AI technologies throughout the entire talent recruitment process, aiming to analyze the role of XAI in enhancing decision-making transparency and traceability. The study finds that XAI, through interpretable algorithms such as LIME, SHAP, knowledge graphs, and causal reasoning, has significantly improved semantic understanding in resume parsing, precision in person-job recommendations, and analytical capabilities in interview evaluations. However, potential biases in generative large models, insufficient cross-scenario interpretability, and computational efficiency issues remain major bottlenecks. Future research should focus on dynamic fairness constraints and the integration of lightweight interpretability tools to promote the coordinated development of XAI in terms of ethical compliance, user trust, and practicality.

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1. Introduction

In the process of talent recruitment, the selection of candidates is of vital importance to enterprises, as it is the core link in implementing talent strategies and optimizing human resource allocation (Ye et al., 2022). The goal of talent recruitment is to identify suitable candidates for specific positions, ensuring that each role is precisely matched with the most appropriate individual. However, traditional recruitment methods rely heavily on the personal knowledge and experience of recruiters, during which subjective factors often lead to decision-making biases. Moreover, differences in experience and qualifications among recruiters further amplify the potential risks of bias.

The development of artificial intelligence (AI) has brought unprecedented opportunities to human resource management and has become the core driving force behind the transformation of the recruitment industry, thereby ushering in a new era of data-driven talent recruitment (Qin et al., 2020). Technologies such as machine learning, deep learning, and large language models have enabled AI to process data with efficiency and accuracy far superior to that of human recruiters, especially in tasks such as screening large volumes of resumes and analyzing job positions. However, these advanced technologies often rely on 'black-box models', which cannot provide interpretable and human-understandable support for the decision-making process (Rudin, 2019). This limitation affects the fairness and transparency of the models. Explainable AI (XAI) aims to provide clear and understandable explanations for black-box models (Zhang & Zhao, 2022).

Explanation is a process that promotes understanding in a way that is comprehensible to humans. Explainability refers to the ability to provide meaning to humans in understandable terms (Arrieta et al., 2020). The concept of XAI was formally proposed as early as 2004 (Van Lent et al., 2004). According to the academic consensus, the definition of XAI can be summarized as providing detailed

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information about the operation mechanisms of artificial intelligence models and the explainability of their results, tailored to different types of human users (Räuker et al., 2023; Wang et al., 2024). In a broad sense, XAI encompasses all technical means that assist users in understanding the behavior of artificial intelligence models. From the developers' perspective, XAI focuses on developing methods, technologies, and tools that can clearly present the decision-making mechanisms of models. This involves designing inherently interpretable models, as well as developing post-hoc explanation techniques to approximate and visualize the behavior of complex 'black-box' models. From the users' perspective, XAI reflects the basis and reasons for model decisions. For example, in recruitment, rejected candidates want to know which résumé attributes most strongly influenced the system's 'not short-listed' verdict so they can judge whether the process was fair and identify concrete improvements for future applications.

XAI enhances transparency through two core paradigms: (1) Intrinsic explainable models (such as clustering, principal component analysis, and decision trees) that directly output interpretable rules; (2) Post-hoc explanation methods (such as Local Interpretable Model-agnostic Explanations (LIME), Partial Dependence Plots (PDP), and Individual Conditional Expectation (ICE)) that infer the logic of black-box models through surrogate models (Liu & Wang, 2024; Wu et al., 2025). XAI technologies can be categorized into three types based on their application stages: (1) Pre-hoc explanation (such as data debiasing preprocessing); (2) In-hoc explanation (such as analysis of linear model coefficients); (3) Post-hoc explanation. In recruitment, post-hoc explanations are more suitable for complex models (such as deep learning). From a decision-making perspective (Kong et al., 2021), XAI can be categorized into: (1) Visual explanations (such as saliency maps); (2) Perturbation-based explanations; (3) Knowledge-based explanations (such as knowledge graphs); (4) Causal explanations (such as Structural Causal Models (SCM) and counterfactual reasoning). Arrieta et al. (2020) systematize XAI along nine dimensions, namely credibility, causality, transferability, informativeness, confidence, fairness, accessibility, interactivity, and privacy awareness. They further partition model transparency into algorithmic transparency, decomposability, and simulatability, and catalogue post-hoc explanations into textual, visual, local, instance-based, simplified, and feature-relevance categories.

The questions this review aims to address are threefold:

- Q1: How can we systematically review the current application status of Explainable Artificial Intelligence based on the different stages of the talent recruitment process?
- Q2: How can the methods or techniques of XAI be embedded into the key stages of talent recruitment?
- Q3: What are the core challenges faced by current research on Explainable Artificial Intelligence in the field of talent recruitment? What key issues should future research directions focus on?

This paper makes the following contributions: (1) It is the first to systematically integrate XAI technology with the entire recruitment process (from job posting to hiring decisions); (2) It proposes a differentiated XAI technology fit framework for each recruitment stage; (3) It reveals the new interpretability challenges caused by generative large models in recruitment (e.g. synthetic data bias), pointing the way for future research. Previous work has investigated the impact of AI on human resource management but has not explicitly integrated it with the recruitment process. This paper provides a detailed overview of the application of explainable AI throughout the recruitment process (Figure 1), identifies the current challenges, and discusses future directions to promote research in this area.

2. Literature review

The integration of AI into societal and organizational domains has attracted extensive scholarly attention, with research examining its implications through diverse methodologies. Bircan and Özbilgin (2025) explore how AI may either perpetuate or alleviate societal inequalities, emphasizing bias and power dynamics. In human resource management (HRM), studies such as Deepa et al. (2024) and Verma et al. (2023) develop integrative frameworks to link AI adoption to organizational performance, employing systematic literature reviews and bibliometric analysis. Lazaroiu and Rogalska (2023) synthesize 2022–2023

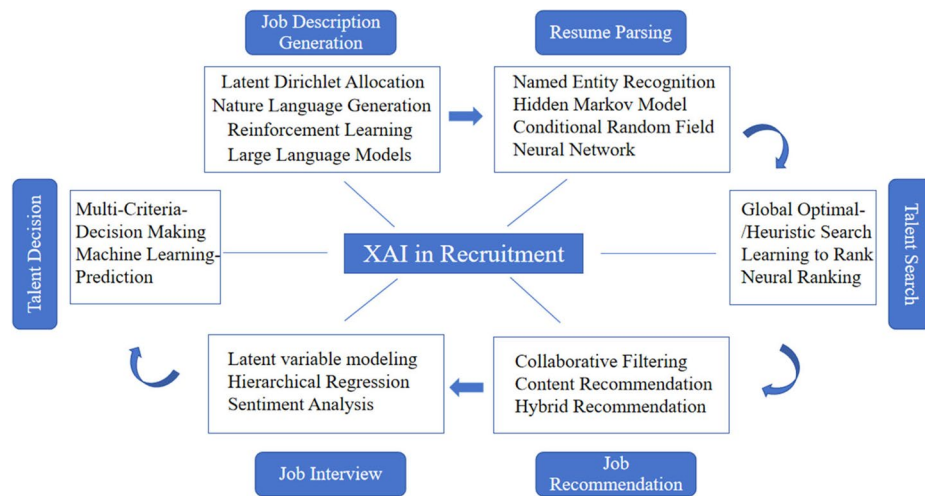


Figure 1. The application of XAI in the entire recruitment process.

literature on generative AI, the metaverse and Web3, systematically mapping applications such as digital twins, 3-D modeling, NLP and computer vision in industry, commerce and labor markets.

The explainability of AI systems constitutes another major research stream. Haque et al. (2023) analyze XAI from an end-user perspective, focusing on trust and transparency, while Liu and Wang (2024) propose a general interpretable machine learning framework for information management. Complementing these theoretical advances, empirical studies like Kim et al. (2023) reveal user preferences for local and visual explanations. In recruitment settings, real-time AI interfaces such as chatbots are shown to influence impression management, sometimes promoting honesty, other times incentivizing deception (Suen and Hung, 2024).

Ethical dimensions are also critically evaluated. Poszler and Lange (2024) review how intelligent decision-support systems shape ethical decision-making, and Salari et al. (2025) synthesize perspectives on AI-induced unemployment. Cramarenco et al. (2023) employed a systematic literature review methodology, integrating the PRISMA flowchart and Khan's five-step framework, to examine the 'skill dimension' and 'well-being dimension', which reveal that AI-driven technological change has rendered continuous reskilling and upskilling an organizational norm.

Recent work by Tang et al. (2025) systematically reviews 85 studies on explainable person-job recommendation systems (PJRS). Their analysis categorizes techniques into data, model, and output layers, featuring methods such as feature attribution, knowledge graphs, and counterfactual explanations, and introduces an integrated framework to enhance transparency.

Despite these advancements, significant limitations and unexplored areas persist in the literature, particularly regarding XAI in recruitment. First, there is a lack of in-depth exploration of the specific technical bottlenecks of XAI from a computer science perspective. Second, existing research has not systematically analyzed the varying technical requirements and explanation needs across the different stages of the recruitment process (e.g. from job posting and resume screening to interview evaluation and final hiring decisions). Finally, the potential pathways for integrating XAI into future recruitment workflows remain underexplored, indicating a critical gap between theoretical frameworks and practical, scalable implementation.

Collectively, these studies employ rigorous methodologies including SLRs, bibliometric analyses, and mixed-methods approaches to address pressing questions related to AI's societal, organizational, ethical, and explanatory challenges, forming a robust yet evolving foundation for future research.

3. Methodology

We conducted an initial scoping search of relevant articles in three research databases: SCOPUS, WOS (Web of Science), and CNKI. An overview of the search process is summarized in Figure 2, with the results presented in Tables 1 and 2. For the English-language databases, we used the following search string: TITLE-ABS-KEY(('Explainable AI' OR 'XAI' OR 'Artificial Intelligence' OR 'Machine Learning' OR 'Deep

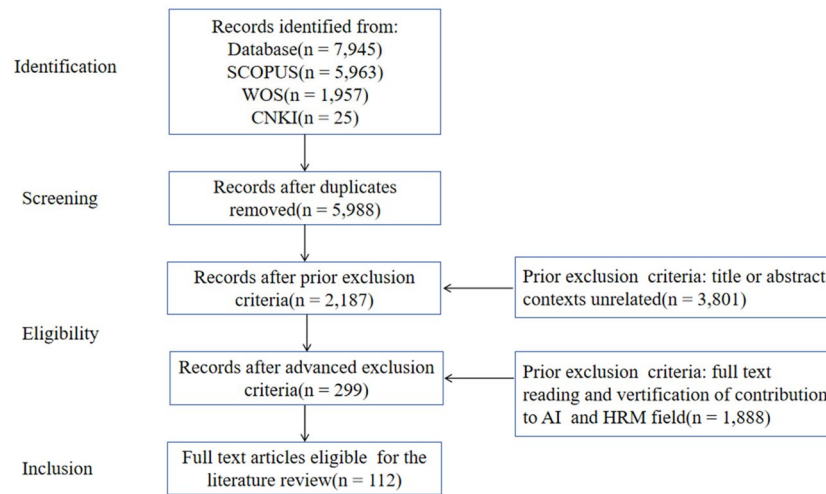


Figure 2. Literature search process.

Table 1. Publications by journals.

Journals	CAS journal ranking	Numbers of articles
ACM Transactions on Information Systems	2	1
ACM Transactions on Intelligent Systems & Technology	3	1
ACM Transactions on Management Information Systems	4	2
Economic Computation & Economic Cybernetics Studies & Research	4	1
Human Resource Development Quarterly	3	1
IEEE Transactions on Knowledge and Data Engineering	1	1
IEEE Transactions on Pattern Analysis and Machine Intelligence	1	1
Information Fusion	1	1
Information Sciences	2	1
International Journal of Advanced Computer Science and Applications	4	2
International Journal of Information and Learning Technology	3	1
International Journal of Knowledge-based and Intelligent Engineering Systems	4	1
International Journal of Organizational Analysis	4	1
Journal of Business Research	1	1
Journal of Electronic Imaging	4	1
Journal of Management Education	3	1
Pattern Recognition	1	1
PeerJ Computer Science	4	1
Peer-to-Peer Networking and Applications	4	1
Total		20

Table 2. Publications by high-quality conferences.

Conferences	Important international conference ranking by tsinghua university	Numbers of conference paper
AAAI Conference on Artificial Intelligence	B	1
Academy of Management Annual Meeting Proceedings	A	1
ACM Designing Interactive Systems Conference	C	1
ACM International Conference on Information & Knowledge Management	B	5
ACM International Conference on Web Search and Data Mining (WSDM '21)	B	1
ACM SIGIR Conference on Research and Development in Information Retrieval	B	1
ACM SIGKDD International Conference on Knowledge Discovery & Data Mining	A	2
Annual Meeting of the Association for Computational Linguistics	A	4
Conference on Empirical Methods in Natural Language Processing	A	4
Conference on Neural Information Processing Systems	A	3
Hawaii International Conference on System Sciences	B	1
IEEE 39th Meeting of the Association for Data Engineering (ICDE)	B	1
IEEE International Conference on Big Data	B	2
IEEE Conference on Computer Vision and Pattern Recognition	A	1
International Conference on Learning Representations	A	2
International Conference on Multimodal Interaction (ICMI '23)	B	1
International Conference on World Wide Web	A	3
International Joint Conferences on Artificial Intelligence	B	1
International Joint Conference on Natural Language Processing	A	1
Total		36

Learning' OR 'LIME' OR 'SHAP') AND ('Talent Search' OR 'Recruitment' OR 'Hiring Decisions' OR 'Human Resource Management' OR 'HRM')). For the Chinese database, we used the following search string: TKA=('可解释' OR '人工智能' OR '机器学习' OR '深度学习' OR 'LIME' OR 'SHAP') AND TKA=('招聘' OR '人力资源管理' OR '面试' OR '人才决策' OR '简历').

Our search included publications written in English or Chinese with no time limit constraints. The inclusion criteria were as follows: (1) Articles published in international peer-reviewed journals and conference proceedings. The filtering mechanism employed the following inclusion criteria: (1) All types of peer-reviewed journal articles to ensure scientific rigor; (2) Articles published only in English or Chinese; (3) The search strings should appear in the article's abstract, title, or author-listed keywords; (4) The selected subject areas are business, management, and computing; (5) Additional keywords related to interpretability methods, such as LIME and SHAP, which may be missing from the search strings but recommended by the search interface, were also included.

After the above screening, we further categorized the literature by theme to facilitate subsequent analysis. Each theme corresponds to technologies used in a recruitment stage and the application of XAI within it.

4. Findings: a stage-based review of XAI in talent recruitment

This section presents the findings of our literature review, structured along the six key stages of the talent recruitment process: job description generation, resume parsing, talent search, job recommendation, job interview, and talent decision. For each stage (Sections 4.1 to 4.6), this paper first review the prevalent AI technologies employed and then delve into the specific application and challenges of XAI methods within that context. This stage-wise synthesis systematically catalog the current state of XAI applications and elucidate how XAI techniques are embedded into each process.

4.1. Job description generation

Job Description Generation (JDG) is defined as the process of automatically generating industry-compliant job texts based on structured inputs (such as lists of skills and company culture). The core challenge of JDG lies in balancing the completeness of information with the naturalness of language. The process of writing job descriptions is typically cumbersome and costly, often requiring a significant amount of human effort. Therefore, the goal of automatically generating job descriptions is to improve efficiency and reduce labor costs.

4.1.1. Technical review of job description generation

Traditionally, early natural language generation (NLG) methods typically employed template-based and rule-based approaches. However, with the advancement of deep learning technologies, a range of neural network-based models have emerged. Among them, sequence-to-sequence models have become a common choice for text generation, a method first proposed by Sutskever et al. (2014). Subsequent studies introduced variants to improve performance. For instance, Bahdanau et al. (2015) extended sequence-to-sequence models by incorporating attention mechanisms between source and target language words. Xia (2017) proposed a Deliberation Network with bidirectional decoders to enhance neural machine translation. Tu et al. (2016) introduced a coverage-based attention mechanism to address over-generation in text generation. Gu et al. (2016) integrated a copy mechanism to handle rare words by copying source words into appropriate target positions. Dong (2019) validated the effectiveness of pre-trained language models in text generation by combining unidirectional, bidirectional, and sequence-to-sequence unsupervised language modeling objectives. In recent years, reinforcement learning has also been applied to the training of sequence-to-sequence models. This is achieved by optimizing discrete and non-differentiable evaluation metrics during the training process, designing reward functions, and employing policy gradient methods to complete NLG tasks. Ranzato et al. (2016) used policy gradient methods to directly optimize ROUGE and BLEU scores for sequence-to-sequence models in text summarization, machine translation, and image captioning. Li et al. (2016) optimized handcrafted reward functions (e.g. answerability, information flow, semantic coherence) combined with policy gradients to diversify and cohere

dialogue responses. Rennie et al. (2017) introduced self-critical policy gradient training to mitigate high variance in gradient estimation. These methods and their variants have proven effective in image captioning, summarization, and reading comprehension. Actor-critic approaches further reduce variance: Bahdanau et al. (2015) proposed a Q-learning-based actor-critic for machine translation, while Chen, Ding, He, et al. (2018) applied an advantage actor-critic with sentence-level metric rewards for summarization. Recently, Liu, Liu, et al. (2020) developed NlySama, a data-driven two-step neural model for job post generation.

With the rise of large language models, generative AI such as GPT can rapidly produce a job description within seconds based on a prompt, thereby eliminating the need for humans to perform this task entirely (Clavié & Soulié, 2023; DeCorte et al., 2023; Mohapatra & Mohapatra, 2022; Ye et al., 2022). Researchers have also evaluated the bias and realism of generated data. Results indicate that, although diversity-encouraging prompts did not significantly improve bias or enhance realism, fine-tuning could significantly increase the realism of advertisements and reduce bias (Borchers, 2022).

4.1.2. XAI applications in job description generation

Probabilistic topic models align latent themes with human-interpretable concepts, aiding model explainability. For example, Ramage et al. (2009) introduced Labeled-LDA, mapping latent topics to predefined labels for transparent document tagging. McAuley et al. (2015) aligned product ratings with review topics to explain user-item compatibility. Qin et al. (2023) proposed Cajon, an end-to-end capability-aware framework for job description generation, which integrates a novel capability-aware neural topic model to extract skill requirements from recruitment data, coupled with an encoder-decoder network enhanced by capability-focused attention and copy mechanisms. This framework generates comprehensive, representative job descriptions while maintaining interpretability. Similarly, Yan et al. (2019) proposed a Biterm Topic Model that explicitly models co-occurrence patterns across the entire corpus of short texts and demonstrated that this model outperforms LDA on short texts.

Despite XAI advancements, current explainability remains limited, focusing on post-hoc explanations of model decisions rather than transparently elucidating self explanations (i.e. intermediate reasoning steps during generation).

4.2. Resume parsing

Resumes are semi-structured texts, and the organization of their content has distinct hierarchical characteristics. They generally consist of two layers: The first layer includes sections such as personal information, educational background, work experience, project experience, awards, and skills. The second layer comprises the detailed information corresponding to each section (Kopparapu, 2010). Resume understanding can be regarded as a process of information extraction and re-structuring. First, it is necessary to convert resumes into digital text format through technologies such as Optical Character Recognition (OCR) and PDF parsers (Feng, 2025; Yao et al., 2023). Then, text preprocessing is carried out, which mainly involves the normalization of the text, including operations such as noise removal and format unification. Subsequently, the processes of entity extraction, text segmentation, text classification, and clustering are conducted (as shown in Figure 3).

4.2.1. A review of resume information extraction technologies

Early research on resume understanding focused on key entity extraction and semantic structure extraction as core tasks, primarily employing rule-matching methods with strong interpretability. For example, Named Entity Recognition (NER) was used to extract structured entities (such as names, contact information, organizations, dates, etc.) through predefined rule libraries, as well as their positions in the text extracted from various sections (Wang et al., 2024; Zimmermann et al., 2016). Although some studies have generated new training samples by randomly selecting entities with the same length as those in the original dataset from an entity dictionary to replace the entities, thereby improving the generalization ability to different entities to a certain extent (Tang et al., 2025), overall, rule-based and

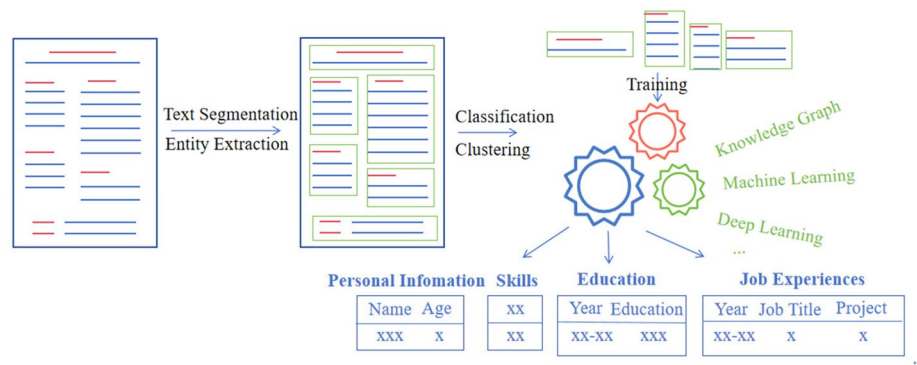


Figure 3. The process of resume parsing.

pattern-matching methods find it difficult to accurately match and generalize in the case of irregular and complex text sequences (Rong, 2024).

Traditional NER methods struggle to capture hierarchical structural features. Therefore, machine learning and deep learning demonstrate better performance in handling hierarchical and sequential text. Some researchers have used Hidden Markov Models (HMM) for text segmentation and label assignment, combined with TF-IDF feature weights to extract serialized information (such as educational experience timelines), and others have employed Support Vector Machine (SVM) classifiers to handle relatively independent modules (such as personal basic information) (Yu et al. 2005; Zhang et al., 2009). Similar methods also include the application of Naive Bayes (NB) classifiers (Bhatia et al., 2019) and the use of Convolutional Neural Networks (CNN) to extract personal information, work experience, skills, educational background, and other information from resumes (Mridha et al., 2021).

With the increasing demand for semantic understanding, machine learning models lack the ability to handle complex semantic relationships and consider text position features. Conditional Random Fields (CRF) models have significantly improved labeling performance by defining composite feature functions. These feature functions not only cover content features (such as word capitalization, punctuation, numbers, etc.) but also integrate layout features (such as font formatting, alignment, line spacing, etc.), thereby enhancing the model's contextual understanding of text sequences (Chen, Gao, & Tang, 2016; Yao et al., 2023). In the field of deep learning, CNN extract local features from vectorized text through multi-scale convolutional kernels. After dimensionality reduction through max-pooling, the multi-channel features are integrated by a merging layer, and the final classification is completed through a fully connected layer (Ayishathahira et al., 2018). Subsequently, researchers have extracted position features and type features corresponding to entities as inputs for Bidirectional Encoder Representations from Transformers (BERT) and used Bidirectional Long Short-Term Memory (BiLSTM) networks to capture the temporal correlation features of text sequences (Zeng et al., 2024; Zhang et al., 2025).

Overall, resume information extraction technologies have undergone a paradigm shift from rule-based to data-driven approaches, with research focus gradually shifting from local feature mining to global semantic understanding. How to improve extraction accuracy while ensuring model robustness, especially adaptability in non-standard expressions and cross-language scenarios, thereby promoting resume parsing technologies towards more practical and generalizable directions, remains a core issue that needs to be urgently addressed.

4.2.2. The application of XAI technologies in resume parsing

In terms of model interpretability itself, there have been efforts to extract modules such as qualifications and skills by constructing data dictionaries, heuristic search, and n-gram pattern matching (Kopparapu, 2010). Additionally, some studies have built ontology knowledge bases by storing domain knowledge related to resumes and generated personal resume ontology files in OWL format to facilitate semantic reasoning (Celik & Elci, 2013). Some research has also measured the impact of each feature on career category classification results through the coefficients of Logistic Regression (LR) models and assessed the contribution of each feature to Random Forest (RF) model predictions using SHAP values (Satheankittiroj & Sa-Ing, 2024).

To address the specific needs of skill analysis, researchers have proposed hierarchical processing methods. For instance, t-distributed Stochastic Neighbor Embedding (t-SNE) dimensionality reduction and clustering have been employed (Zimmermann et al., 2016), along with association rule mining algorithms such as Apriori and FP-Growth to analyze the correlation patterns between ‘hard skills’ and ‘soft skills’ (Li et al., 2023; Ternikov, 2022). Other studies have clustered job seekers and positions separately and then performed weighted similarity matching within these clusters (Rodriguez & Chavez, 2019). Akkasi (2024) combined CRF with BERT and BiLSTM to extract technical/non-technical skills and used LIME to evaluate the contribution of words to label predictions.

Regarding post-hoc explanation, some studies have modeled the semantics of job text using embedding models such as Sent2vec and SBERT, combined with attention mechanisms to quantify sentence importance, and ultimately mapped the results to a structured Competency Map (CMAP) (Ayishathahira et al., 2018; Guignani & Misra, 2020; Khaouja et al., 2021; Ren & Gong, 2025; Wang et al., 2021, 2019; Zhang et al., 2023). Other research has used the LIME method to visualize the contribution of each word in review text to the model’s final classification results through the magnitude of linear model coefficients. For image modality, the Grad-CAM method has been employed to display key areas in the input image that influence the output decision through heatmaps (Yang & Zhang, 2025). It is evident that current deep learning models can only enhance interpretability through proxy model explanations, knowledge graph integration of semantic information, and visualization of intermediate layer weights.

4.3. Talent search

Talent search is a process of information retrieval and ranking, with the goal of returning a relevance-ordered list from a collection of documents based on user queries, enabling users to quickly and accurately obtain the information they need. In the context of talent search and resume screening, the precision and efficiency of ranking search directly affect recruitment efficiency and the quality of talent matching. Moreover, the explainability of talent screening in the recruitment process is crucial for transparent screening mechanisms and mitigating resume bias.

4.3.1. Review of information retrieval and ranking techniques

Based on the differences in the search methods for feature subsets, search strategies can be categorized into three major types: global optimal search, sequential search, and random search (Somol et al., 2004; Sun et al., 2004). Liu et al. (2025) and Khurana et al. (2016) proposed a hierarchical greedy search, which supports depth-first, breadth-first, and balanced search strategies that prioritize breadth before depth. This method constructs a feature generation tree with the original dataset as the root node and calculates the performance of the machine learning model at the root node. Su et al. (2022) introduced a knowledge-enhanced search result diversification method called KEDIV, which leverages entities and relationships from a knowledge base to improve the diversity of search results and the richness of subtopics (Su et al., 2022). Apatean et al. (2017) used machine learning classification methods such as Linear Discriminant Analysis (LDA), K-Nearest Neighbors (KNN), Decision Trees (Tree J48), and Sequential Minimal Optimization (SMO) to classify talent attributes. Liu, Yuan, et al. (2020) quantified user attributes and used k-means clustering to group user attributes. They then calculated the Pearson similarity within the category of the target user to identify a set of neighboring users for the target user.

Traditional search engine ranking algorithms often focus solely on relevance when processing user queries. For instance, models such as inverted indexing, TF-IDF, and BM25, which primarily rely on simple statistical features like term frequency and document length, offer good interpretability. However, with the development of deep learning technologies, an increasing number of researchers have begun to employ complex neural network models to build Information Retrieval (IR) systems in order to enhance retrieval and ranking performance. The decision-making processes of these models, however, are often difficult to understand and interpret. Verma and Ganguly (2019) proposed the Local Interpretation Model for IR (LIRME) based on LIME, which is used to explain the ranking decisions of IR models for specific query-document pairs. Research has shown that the diversity of sampling is crucial for training effective local interpretation models, and the stability of the model is inversely proportional to the number of

parameters used for interpretation. Zhuang et al. (2021) applied Generalized Additive Models (GAMs) to ranking tasks and proposed Neural Ranking GAMs, which can achieve predictive performance comparable to complex ‘black-box’ models while maintaining interpretability. Heuss et al. (2024) conducted research on feature attribution explanations for ranking models and proposed the RankingSHAP method. Singh and Anand (2020) used simple rankers to approximate the original ranking of complex black-box models. Hu et al. (2018) proposed the Search Session Markov Decision Process (SSMDP), which outperforms traditional Learning to Rank (LTR) methods.

Search and ranking together form the process of locating targets from massive data and arranging them according to needs. The search strategy directly affects the upper limit of ranking quality, and the requirements of the ranking model, in turn, affect the optimization direction of the search space. In the future, with the increasing demand for multimodal data processing, the collaborative mechanisms of search and ranking may further integrate causal inference and active learning paradigms, forming a more adaptive intelligent retrieval system.

4.3.2. The application and challenges of XAI technologies in talent search

In resume ranking applications, Manad et al. (2018) combined the occurrence of skills within the document to achieve resume ranking through the calculation of static and dynamic skill scores. Geyik et al. (2018) considered attributes such as work experience and qualifications, using LDA topic modeling to segment the candidate space into multiple intent clusters, each with representative candidate group feature keywords, and then used the Multi-Armed Bandit (MAB) algorithm to select the optimal intent cluster. Ozcaglar et al. (2019) proposed a two-level ranking system, with Gradient Boosting Decision Trees (GBDT) and tree interaction features as inputs to train a Generalized Linear Mixed Model (GLMix) for real-time retrieval and ranking. Other studies have ranked resume features using algorithms provided by Weka software, such as information gain and gain ratio, finding that job titles, work experience, educational background, and marital status ranked the highest, while age and gender ranked lower (Rodriguez & Chavez, 2019).

It can be seen that XAI technologies have enhanced the transparency of model decision-making to some extent but have also revealed the contradiction between explainability and model performance trade-offs. Existing methods are still limited by the adaptability of global explanations to individual-level decision logic and lack real-time response capabilities to bias evolution in dynamic recruitment scenarios. The focus of the solution lies in establishing an effective balance between controlling the granularity of feature attribution and ensuring fairness.

4.4. Job recommendation

The goal of person-job recommendation is to achieve a precise match between individuals and job positions. In recent years, recommendation systems have undergone a significant transformation from traditional recommendation algorithms to deep learning-based recommendation systems. This evolution enables them to not only process surface-level data but also learn the latent feature representations of user behavior. As a result, they have overcome challenges encountered by traditional recommendation systems, such as cold start and matrix sparsity, thereby realizing a qualitative leap in recommendation accuracy. However, the lack of interpretability remains a major existing challenge (Malinowski et al., 2006; Qin et al., 2020).

4.4.1. Review of the basic types and related technologies

In the context of person-job recommendation, there are four common types of recommendation systems: collaborative filtering-based recommendation systems, content-based recommendation systems, hybrid recommendation systems, and recommendation systems based on large language models.

Collaborative filtering-based models focus on extracting preference information from the interaction history between job seekers and positions, and then recommend similar positions based on this information. Some studies have employed topic models or deep learning models to extract user preferences and auxiliary features from reviews, thereby achieving rating prediction (Cheng, Ding, He, et al., 2018; Cheng, Ding, Zhu, et al., 2018).

Content-based recommendation systems recommend new products that have a high similarity to the products in the user's historical interaction records. For resume recommendation, machine learning methods such as K-Means++ and GBDT have been employed to address the resume classification problem (Ozcaglar et al., 2019). Additionally, Qin et al. (2020) proposed an end-to-end bidirectional LSTM neural network model to construct word-level semantic representations of job requirements and candidate experience. Huang et al., (2023) introduced a two-stage resume recommendation model based on deep learning and attention mechanisms, particularly considering the latent preference information hidden in the resumes of hired employees.

Hybrid recommendation integrates collaborative filtering-based recommendation and content-based recommendation, thereby mitigating the limitations of a single paradigm to some extent. Hu et al. (2023) proposed a reciprocal recommendation method that takes into account the mutual job preferences of job seekers and recruiters in the online labor market. Saito and Sugiyama (2024) introduced a job recommendation method (JME) that utilizes both auxiliary behavioral data and explicit preference data of each user, addressing the issue of overlapping behaviors and improving latent representations. Integrated auxiliary behavior and explicit preference data to improve latent representations.

Recommendations based on LLMs can leverage their rich external knowledge, as well as their powerful text processing and reasoning capabilities, to generate more accurate job recommendations. For example, using LLMs in combination with Retrieval-Augmented Generation (RAG) to enhance job recommendation performance through generated job postings has been explored (Du et al., 2024; Lacic et al., 2020; Ntioudis et al., 2022; Zheng et al., 2023). However, its high computational cost and privacy risks remain limitations.

4.4.2. XAI applications in job recommendation

Most interpretable recommendation systems explain the results of recommendations through various machine learning models, knowledge graphs, or textual and visual artifacts (Ribeiro et al., 2016; Song et al., 2019; Zhang et al., 2021; Zhang & Xu, 2018). Upadhyay et al. (2021) also modeled resume texts using hierarchical attention mechanisms and provided both structured and unstructured levels of recommendation explanations. The focus of XAI is not only to make the recommendation process traceable but also to ensure that recommendations are subject to constraints and preferences imposed by scope, rules, and architecture, thereby guaranteeing alignment with job seekers' preferences and organizational requirements.

4.5. Job interview

Interviews are a crucial part of the recruitment process, which can be categorized into behavioral and situational interviews. Behavioral interviews mainly use a series of standardized questions aimed at understanding the candidate's past experiences in handling situations that may arise in future work. Situational interviews focus on constructing hypothetical, future-oriented questions to assess the candidate's ability to respond to similar situations they may face in their future role (Guo & He, 2004). Traditional interviews are largely subject to human subjective factors and are often costly and inefficient. With the development of artificial intelligence, more and more companies are using AI to assist in recruitment activities (Gao & Qiao, 2024), which helps in quickly screening candidates, determining qualifications, scheduling interviews, answering common questions, evaluating feedback, and responding to unsuccessful candidates, reducing the likelihood of losing qualified candidates due to long recruitment cycles (Nawaz & Mary, 2019; Paramita et al., 2024). Especially with the advancement of generative artificial intelligence (GAI), its potential to integrate different human resource development activities has increased (Korzynski et al., 2024). Sonawane et al. (2023) developed a text-based sentiment analysis engine and a customer interview chatbot 'vRecruit' to maintain contact with candidates. MetaVerse technology is also being used for virtual meetings and recruitment interviews, providing employees with an immersive interactive environment (Aydin et al., 2024). Qin (2021) constructed a neural network-based generative model to generate skill-related interview questions and, using click data from search logs, designed an interview question retrieval system that can provide personalized question recommendations for interviewers. Additionally,

a graph-enhanced question retrieval algorithm was designed to efficiently recommend matching interview questions when interviewers query a specific set of skills.

Research results indicate that in terms of applicants' perception of interactional fairness, designing the external image and language style of AI interviewers to highlight their warm and friendly personality traits can effectively enhance interpersonal fairness within interactional fairness; designing AI interviewers to provide high-informativeness feedback can effectively enhance informational fairness within interactional fairness (Wang, 2022). HR professionals generally believe that AI recruitment software can effectively automate and scale existing recruitment processes, thereby helping companies screen candidates from a broader and more diverse talent pool, but there are still concerns due to a lack of trust in the accuracy of data and limited control over the algorithmic matching process (Agnihotri et al., 2023; Li et al., 2021; Vedapradha et al., 2024). Vedapradha et al. (2023) collected questionnaire rating data from 384 HR and talent acquisition managers (including variables such as actual usage, perceived usefulness, perceived ease of use, and talent management, etc.), and used multiple linear regression analysis to conclude that the higher the level of understanding of AI technology, the higher its adoption rate in the talent recruitment process. As a key factor in the digital transformation of human resource management, explainability is a core element in the use of AI models.

4.6. Talent decision

In the process of talent recruitment and decision-making for the optimal plan, Multi-Criteria Decision-Making (MCDM) methods are widely employed. MCDM methods balance the subjective perceptions and objective benchmarks in the talent identification process and can be used to aggregate individual judgments of group members into a single, representative group judgment or to construct group choices from individual selections (Saaty, 2008).

Department managers or chief examiners tend to rigorously evaluate job applicants from multiple aspects such as educational background, ability, quality, and age. Given the limited time and insufficient information, chief examiners often conduct pairwise comparisons between any two candidates. A decision-making method based on consistency analysis has been proposed, which helps to check whether the preferences provided by department managers are self-contradictory (Xiao et al., 2022; Ye et al., 2022). Building on the MCDM framework, some studies have combined theories such as T-spherical fuzzy sets (T-SFS) and entropy weights (Ji et al., 2021; Liao et al., 2019; Yan et al., 2024; Zafar et al., 2021; Zhang et al., 2024). Lai and Ishizaka (2020) integrated AHP and FlowSort to identify talents and provided employees with intuitive feedback through the visualization tool GAIA. Other research has employed machine learning methods such as Logistic regression, SVM, KNN, DT, and Multilayer Perceptron (MLP) to predict the future performance of candidates (Demircan & Aksac, 2022; Sonawane et al., 2023).

However, in AI-supported recruitment processes, biases may infiltrate datasets and influence the decision-making of AI algorithms, potentially exacerbating disparities among minority groups and gender gaps (Simon et al., 2023). Zhang (2020) demonstrated through regression analysis that when job applicants possess traits that conform to gender stereotypes, interviewers perceive a higher fit and provide better interview evaluations. Moreover, the impact of gender stereotypes on perceived job fit is smaller when interviewers have a stronger proactive personality.

5. Discussion

Based on the comprehensive review of AI and XAI applications across the recruitment pipeline presented in Section 4, this discussion section synthesizes these findings to address their broader implications and challenges. The core AI technologies discussed across all six stages are summarized in Table 3, which provides a concise overview of the technical evolution in this domain. And a comparative framework consolidating the XAI methods reported in Sections 4 is presented in Table 4, detailing their respective forms, advantages, and limitations across recruitment scenarios. Subsequently, we critically analyze the core challenges faced by current research and, building upon this analysis, propose targeted future research directions.

Table 3. Technological development process in the entire talent recruitment process.

Application scenario	Technological development process	Authors and year
Job description generation	Sequence-to-Sequence models and their variants	Bahdanau et al. (2015); Dong (2019); Gu et al. (2016); Sutskever et al. (2014); Tu et al. (2016); Xia (2017)
	Reinforcement Learning	Bahdanau et al. (2015); Li et al. (2016); Liu, Yuan, et al. (2020); Ranzato et al. (2016); Rennie et al. (2017)
	Large Language Models	Clavié and Soulié (2023); DeCorte et al. (2023); Joshi et al. (2018); Liu and Ge (2022); Mohapatra and Mohapatra (2022); Ye et al. (2022)
Resume parsing	Entity extraction	Zimmermann et al. (2016); Wang et al. (2024); Tang et al. (2025); Rong (2024); Yu et al. (2005); Zhang et al. (2009); Bhatia et al. (2019); Mridha et al. (2021); Zeng et al. (2024)
	Text segmentation	Yao et al. (2023); Chen et al. (2016); Ayishathahira et al. (2018)
Talent search	Classification and clustering	Zhang et al. (2025)
	Genetic algorithms and bayesian optimization	Li et al. (2024)
	Hierarchical greedy search	Khurana et al. (2016)
	Knowledge enhancement	Su et al. (2022)
	Classification(LDA、 KNN、 Tree J48、 SMO)	Apatean et al. (2017)
	Clustering(K-means)	Liu, Yuan, et al. (2020)
	Reinforcement Learning	Hu et al. (2018)
Job recommendation	Combinatorial optimization	Singh and Anand (2020)
	Collaborative filtering	Cheng, Ding, He, et al. (2018); Cheng, Ding, Zhu, et al. (2018)
	Content recommendation	Huang et al. (2023); Ozcaglar et al. (2019)
	Hybrid recommendation	Hu et al. (2023); Qin et al. (2020); Saito and Sugiyama (2024)
	Large Language Models	Zheng et al. (2023)
	Graph enhancement	Qin (2021)
	Regression analysis	Vedapradha et al. (2023)
Talent decision	Sentiment analysis, Generative AI	Sonawane et al. (2023); Tay et al. (2024); Venkatachalam and Venkatachalam (2023); Aydin et al. (2024)
	Multi-Criteria decision analysis	Demircan and Aksac (2022); Ji et al. (2021); Lai and Ishizaka (2020); Liao et al. (2019); Yan et al. (2024); Zhang et al. (2024); Zafar et al. (2021)
	Classification and regression	Demircan and Aksac (2022); Sonawane et al. (2023); Zhang (2020)
	Decision support techniques	Liao et al. (2019); Xiao et al. (2022); Ye, Zhan, et al. (2022); Zafar et al. (2021)

5.1. Summary of XAI methods

The interpretability methods used in the aforementioned application scenarios, along with their corresponding forms of explanation and main advantages and disadvantages, are summarized, which encompass interpretable models, the interpretability of internal model components, and post-hoc explanations (Table 4).

5.2. Future research directions

Based on the limitations of existing research, future efforts can focus on the following directions to deepen exploration and promote the transparency, credibility, and practicality of XAI in recruitment.

5.2.1. Knowledge-Enhanced job description generation

In the area of job description generation, creating job requirements not only necessitates the production of overall coherent text but also requires the precise inclusion of key content such as specific skills. However, long-text-to-long-text generation is prone to information loss (Liu, Yuan, et al., 2020). Future research should focus on enhancing the accuracy and interpretability of generated text. It is essential to explore methods that integrate knowledge graphs and neural topic models. By explicitly modeling the relationship between skills and job requirements, more interpretable job descriptions can be generated. Additionally, investigating how to reduce bias in generated text through prompt engineering and fine-tuning techniques is also an important direction.

5.2.2. Multimodal resume parsing and debiasing

In the area of resume understanding, current research rarely involves the study of resume language style. The language style of a resume may contain potentially important information about the job seeker's

Table 4. Comparison of the application of explainable methods in the entire talent recruitment process.

Application scenario	Form of explanation	Main advantages	Main disadvantages	Authors and year
Job description generation	Explicit structure	Associating topics with tags to make the model's results more understandable and interpretable.	Reliant on high-quality labeled data. The correspondence between topics and tags may be overly simplistic.	Ramage et al. (2009); Yan et al. (2019); McAuley et al. (2015)
	Feature attribution	By examining attention weights, gain a better understanding of the model's decision-making process.	Understanding the meaning behind the weights is difficult. The attention mechanism may introduce noise.	Qin et al. (2023)
Resume parsing	Explicit structure	Global explanation	Clustering results are sensitive to the chosen clustering algorithm and parameters.	Celik and Elci (2013); Kopparapu (2010); Li et al. (2023); Rodriguez and Chavez (2019); Ternikov (2022);
	Post-hoc explanation	Visual explanation	Gradient vanishing. Sensitive to noise.	Satheankittiroj and Sa-Ing (2024); Yang and Zhang (2025); Zimmermann et al. (2016)
	Feature attribution	Providing probabilistic outputs, which can help users understand the model's confidence in different labels.	High training cost. Inherently complex. Understanding the meaning behind the weights is challenging. The attention mechanism might introduce noise.	Akkasi (2024); Ayishathahira et al. (2018); Gughani and Misra (2020); Khaouja et al. (2021); Ren and Gong (2025); Wang et al. (2021); Zhang et al. (2023)
Talent search	Explicit structure	Easy-to-understand topics. Viewing the distribution of documents across various topics.	The quality of topics depends on the data.	Geyik et al. (2018); Manad et al. (2018)
	Post-hoc explanation	Global consistency.	High computational complexity. Assumption of feature independence.	Parasurama and Sedoc (2021); Sogancioglu et al. (2023)
	Feature attribution	Global explanation.	Sensitivity to feature selection. Overfitting.	Ozcaglar et al. (2019); Rodriguez and Chavez (2019)
Job recommendation	Feature attribution	By examining attention weights, one can better comprehend the decision-making process of the model.	It is challenging to understand the implications behind the weights. The attention mechanism may introduce noise.	Schellingerhout (2024); Upadhyay et al. (2021); Zhang and Xu (2018); Zhang et al. (2021)
	Interactive explanation	High flexibility.	Lack of structured information.	Ribeiro et al. (2016)
Job interview	Explicit structure	Strong explainability. High computational efficiency. Wide applicability.	Linear assumption. Lack of causal explanation.	Gao and Qiao (2024)
	Interactive explanation	Active learning. Interactive explanation.	Complex design. Feedback bias. Poor scalability.	Wang (2022)
Talent decision	Explicit structure	Global interpretability. Simple and intuitive.	May not be highly accurate. Can only capture linear relationships.	Demircan and Aksac (2022); Sonawane et al. (2023); Zhang (2020)
	Explicit structure	Structured decision-making. Quantification of subjective judgments. Consistency testing.	Strong Subjectivity. Scale limitations. Difficult to handle complex dependencies.	Lai and Ishizaka (2020)
	Feature attribution	Local interpretability. Model-agnostic.	High computational cost. Inability to explain complex nonlinear relationships.	Ji et al. (2021); Liao et al. (2019); Zhang et al. (2024)

personality and communication skills, which has not yet been fully explored and utilized. Future work could combine NLP and sentiment analysis techniques to deeply mine information about language style, personality traits, and communication skills in resumes. Meanwhile, explanations for this information could be provided through XAI technologies. In addition, multimodal resume parsing (such as combining text, images, and videos) is also a direction worth exploring.

In the area of talent search, despite significant advancements in semantic search and personalized recommendation technologies, issues of resume bias and fairness persist (Liu et al., 2023).

Parasurama and Sedoc (2022) compared the gender incongruity scores calculated using the Tf-Idf+Logistic model, the Longformer model, and the agentive and communal word dictionaries in LIWC, and concluded that gender incongruity has a negative impact on women's recruitment outcomes. Additionally, they found that gender-neutral word embedding models have limited effectiveness in concealing gender information, and it is necessary to balance debiasing with algorithmic performance (Parasurama & Sedoc, 2021). Sogancioglu et al. (2023) employed the bias-mitigation algorithm ProxyMute, which leverages the global feature importance identified by SHAP, to recognize and disable proxy features on two datasets containing sensitive attributes such as gender, age, and race, thereby reducing their influence on recruitment decisions. Wu and Ye (2025) proposed a regularization-based training method that introduces a debiasing loss (Debias Loss) to mitigate gender bias in large language models during the resume screening task.

Future research could focus on how to detect and eliminate bias within search algorithms using XAI technologies, such as demonstrating the impact of different features on search results through counterfactual explanations. Additionally, developing search algorithms that are cross-cultural and cross-lingual, ensuring their fairness and explainability across different contexts, is also an important research direction.

5.2.3. Causal reasoning-empowered job recommendation

In the area of person-job recommendation, since large models are trained on datasets that may contain biases (Gallegos et al., 2024), the system may perpetuate these biases, thereby affecting the fairness, neutrality, and interpretability of the generated job postings.

Future research could explore how to enhance the interpretability of LLM through prompt engineering, model distillation techniques, and causal graphs such as reasoning graphs for LLM (Wang et al., 2024), and provide clear reasons for recommendations to HR professionals. Companies can integrate lightweight tools by developing API interfaces or plugins to embed XAI modules into existing recruitment systems. Additionally, scenario-based explanation adaptation is a practical concern for HR, which involves dynamically adjusting the complexity of explanations according to job types. For example, for technical positions, detailed algorithmic weights for skill matching can be displayed, while for administrative positions, the evaluation logic for communication skills can be emphasized. Throughout the process, managers should be allowed to question model suggestions and provide feedback to correct the model, thereby achieving an adaptive feedback loop.

5.2.4. Multimodal explainable interview assessment

Individuals from different backgrounds (such as data scientists, product managers, and clients) have varying levels of understanding and needs regarding the technical details of AI models. Gao and Qiao (2024) categorized the transparency of AI interviews into procedural interaction transparency and decision rule transparency. Through hierarchical regression, they found that procedural interaction transparency has a significant positive correlation with the applicants' sense of fairness, while decision rule transparency has a significant negative correlation with the sense of fairness. Therefore, it is necessary to strike a balance between technical details and business needs, avoiding information overload while ensuring that explanations support the credibility and acceptability of the model (Dhanorkar et al., 2021). For HR professionals, XAI tools available on the market, such as HireVue, XOR, ParadoxHiretual, Eight times, Pymetrics, and Eightfold, utilize methods like natural language processing and virtual reality to analyze interview videos or audio data, providing an objective supplement to interview assessments and significantly reducing the workload of HR tasks (Tay et al. 2024; Venkatachalam & Venkatachalam, 2023). On one hand, by establishing models and training data, artificial intelligence can provide accurate assessment results during the automated interview process, aiding recruiters in making better decisions.

In the area of interview evaluation, existing research has primarily focused on the development of automated interview tools, which, however, lack transparency in their decision-making processes. Future work could involve integrating multimodal data to develop interpretable interview assessment models. These models could utilize techniques like heatmaps and attention mechanisms to highlight the key

bases for interview scoring. Additionally, hierarchical explanation permissions could be implemented. This approach would provide managers with detailed technical explanations, while candidates would only be shown summary-level results. In this way, transparency can be ensured while also protecting privacy, which is an important direction for future development.

5.2.5. Transparent talent decision

Lastly, in the area of talent decision, although MCDM methods have been widely applied, their complexity and subjectivity limit transparency. Future exploration could focus on how to simplify the decision-making process through XAI technologies and provide HR with clear decision paths and bases. Additionally, researching how to introduce fairness constraints in large-scale talent decision-making, reduce algorithmic bias, and enhance the transparency and consistency of decisions through visualization tools is also an important direction for future research.

In the systematic construction of the XAI framework, ethics, fairness, and accountability are core principles. First, ethics require that system design adhere to principles of privacy protection and data security to ensure that candidates' personal information is not improperly used or abused. Human elements are particularly important in recruitment, as candidates often expect timely feedback and interaction with recruiters during the application process. Respect for candidates' personalities and emotions makes the recruitment process more comprehensive and people-oriented. Second, fairness requires the system to ensure predictive consistency across different groups through regular audits and counterfactual explanations, and to proactively identify and correct historical biases in the algorithms, such as reliance on unnecessary features. Lastly, accountability requires the system to record all causal pathways of decisions to trace responsibility and enhance decision explainability. Additionally, user education is a key part of the framework's implementation, and it is necessary to help HR understand the limitations of XAI through simulated training, strengthen human-led human-machine collaboration, avoid over-reliance on technical outputs, and ensure that HR makes the final judgment in the interview decision-making phase, not AI.

In summary, this review systematically embeds XAI technologies into the entire talent recruitment process for the first time, revealing the dual potential of AI in improving recruitment efficiency and fairness. The article chose to exclude some literature that might provide additional insights. For example, what forms of explanation can truly be trusted by HR or job seekers? From the user's psychological mechanism, which level of explainability is more preferred? How can XAI technologies have practical cognitive utility for job seekers or HR and achieve a balance with ethical constraints and legal regulations? Therefore, the limitation lies in staying mostly at the algorithm level, lacking interdisciplinary experimental design, such as not starting from the actual situation of enterprises. For example, when dealing with a resume database of millions, the computational delay of local explanation models may offset the efficiency advantages, and this key contradiction has not been included in the discussion of technology selection. How to define responsibility when XAI's decision recommendations conflict with HR experience? At the intersection of algorithms and humanity, what we need is not only smarter machines but also a wiser collaborative paradigm. Future research should focus on enhancing the transparency, fairness, and practicality of XAI in the entire recruitment process, while also paying attention to ethical, legal, and user acceptance issues. Through multi-dimensional design to achieve fairness and traceability in the recruitment process, while strengthening user trust; with technological innovation and interdisciplinary research, XAI is expected to make greater breakthroughs in improving recruitment efficiency, accuracy, and fairness, creating greater value for enterprises and candidates.

Registration and protocol

This review was not registered, and no formal protocol was prepared. Amendments: No amendments were made, as no protocol or registration was in place.

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Author contributions

CRedit: **Gening Zhang**: Data curation, Visualization, Writing – original draft, Writing – review & editing; **Lin Pan**: Formal analysis, Methodology, Writing – original draft, Writing – review & editing; **Fang Tang**: Conceptualization, Methodology, Supervision, Validation; **Feng Yao**: Project administration, Supervision, Validation.

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Data availability statement

The data that support the findings of this study are available from the author, Gening Zhang, upon reasonable request.

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